

A Review of Agent-Based Programming for Multi-Agent Systems

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1. RESEARCH OBJECTIVE & SYSTEM ARCHITECTURE

Surveys agent programming languages categorized by BDI (Belief-Desire-Intention) models, logic-based engines, and speech-act message passing (FIPA-ACL).

2. BENCHMARK DATASETS & EMPIRICAL EVALUATION

Tooling and programming language corpus (Jason, JADE, GOAL, Astra, 2APL).

Key Numerical Findings & Empirical Proofs:

Standardizes agent coordination taxonomy and formal semantics for autonomous agents operating in multi-agent environments.

3. CRITICAL ARCHITECTURAL FLAWS & METHODOLOGICAL LIMITATIONS

Identified System Failure Modes:

Rigid deterministic boolean pre-conditions; completely incapable of handling probabilistic ambiguity or generative reasoning; no confidence estimation.

4. RESEARCH GAP & RELEVANCE TO CAG-DELPHI

This work exemplifies the *Unconditional Consultation Dilemma* observed across the 15-paper corpus. By failing to incorporate epistemic uncertainty self-gating prior to multi-agent activation, the architecture incurs severe token inflation and exposes confident primary solutions to debate degeneration noise. In CAG-Delphi, this paper serves as an empirical anchor for diversity-governed consensus and comparative baseline evaluation.

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